

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\theta(t_1 - t_2) \begin{bmatrix} 0 & \frac{\theta(-n-\frac{1}{2})t_1^n}{t_2^{n+1}} \\ \frac{\theta(n+\frac{1}{2})t_2^n}{t_1^{n+1}} & 0 \end{bmatrix} - \theta(t_2 - t_1) \begin{bmatrix} 0 & \frac{\theta(n+\frac{1}{2})t_1^n}{t_2^{n+1}} \\ \frac{\theta(-n-\frac{1}{2})t_2^n}{t_1^{n+1}} & 0 \end{bmatrix}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} (\sigma \circ (\sigma \circ f), M, ((\sigma \circ f)^* h) \circ (f^* h \circ \phi), t) &= (f, M, (f^* \sigma^* h) \circ f^* h \circ \phi, t) \\ &= (f, M, f^*(\sigma^* h \circ h) \circ \phi, t) \\ &= (f, M, f^*(\text{Id}_L) \circ \phi, t), \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} &N_{L_n}^d \mathbb{E} \left(\text{Tr}(E_{H_{C_p}^\omega}(J_{L_n, E})) (\text{Tr}(E_{H_{C_p}^\omega}(J_{L_n, E}) - 1) \right) \\ &\leq N_{L_n}^d \left(Q_\mu(|J_{L_n, E}|) |C_p| \right)^2 \\ &\leq N_{L_n}^d (|J_{L_n, E}|^\alpha |C_p|) ^2 \\ &= O \left(\beta_{L_n}^{-2\alpha} N_{L_n}^d \left(\frac{2L_n + 1}{N_{L_n}} \right)^{2d} \right). \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\int\limits_0^t \int\limits_{\mathbb{R}^3} -v \cdot \vec{\varphi}_2 - \frac{1}{k_1 \varrho} \nabla \xi \cdot \vec{\varphi}_2 \, \mathrm{d}x \mathrm{d}s = \tau^2 \int\limits_0^t \int\limits_{\mathbb{R}^3} v_t \cdot \vec{\varphi}_2 + k_1 v \cdot \nabla v \cdot \vec{\varphi}_2 \, \mathrm{d}x \mathrm{d}s.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} E_1 &= \det T(\phi)T(\phi_R)^{-1} \cdots T(\phi_1)^{-1}T(\psi_S)^{-1} \cdots T(\psi_1)^{-1}T(\psi_0)^{-1} \\ E_2 &= \det T(\tilde{\phi})T(\tilde{\phi}_R)^{-1} \cdots T(\tilde{\phi}_1)^{-1}T(\tilde{\psi}_S)^{-1} \cdots T(\tilde{\psi}_1)^{-1}T(\tilde{\psi}_0)^{-1} \\ E_3 &= \prod_{j=0}^S \det T(\psi_j)T(\psi_j^{-1}). \end{aligned}$$